Q12 — Deeper projects

Why do models give confident answers even when wrong?

Deep Learning Systems 2026 — AI Internals Module

Framing the question

Confident wrong answers are a symptom, not a single mechanism. They arise from the interaction of several internal properties: modern neural networks are *poorly calibrated*, with confidence systematically exceeding accuracy [2]; maximum-likelihood training pushes probability mass toward sharp distributions regardless of epistemic uncertainty; and reinforcement learning from human feedback can reward assertive, fluent responses over hedged ones, amplifying overconfidence and hallucination [3, 6]. A further source is that language models can be "calibrated" at the token level yet still produce factually wrong claims stated fluently, because fluency and factuality are governed by different internal pathways [4]. The projects below ask you to measure, localize, and partially mitigate these phenomena, treating confidence as an internal computation that can be probed rather than an opaque model trait.

Project 1 — Calibration cartography across tasks and scale

Goal. Build an empirical map of how calibration error relates to model scale, task type, and decoding temperature, and determine whether overconfidence is uniform or concentrated in specific regimes.

Approach. Select at least three model sizes from an open family. Evaluate on a mix of multiple-choice benchmarks (where token probabilities over answer letters give a clean confidence signal) and short free-form QA (where you elicit a verbalized confidence or use sequence likelihood). Compute expected calibration error (ECE), Brier score, and reliability diagrams following Guo et al. [2], and separately measure selective-prediction quality via risk–coverage curves. Contrast pretrained checkpoints against instruction-tuned ones to test whether alignment tuning degrades calibration [4, 6].

Deliverables.

- Reliability diagrams and ECE/Brier tables broken down by model size, task type, and temperature.
- A risk–coverage analysis with at least two selection scores.
- A 4–6 page report stating where overconfidence is worst and whether it is driven by scale, tuning, or decoding, with reproducible code.

Project 2 — Does the model "know" it is wrong internally?

Goal. Test whether a model's hidden states encode a usable signal of its own correctness that is stronger than its stated confidence.

Approach. Collect model answers labeled correct/incorrect on a factual-QA set. Train lightweight linear probes on frozen hidden representations at several layers to predict correctness, following the self-knowledge paradigm of Kadavath et al. [4] and probing methodology from Alain and Bengio [1]. Compare probe AUROC against the model's own token-level or verbalized confidence. Investigate layer-wise emergence: does an internal correctness signal appear before the final layer, and is it discarded by the output head? Optionally relate findings to representation-level factuality signals in the hallucination literature [3].

Deliverables.

- Layer-wise probe AUROC curves vs. baseline confidence.
- An analysis of whether the "knows-it-is-wrong" signal is linearly decodable and where it peaks.
- A short report discussing implications for why the output layer can be confident when intermediate states are not.

Project 3 — Mitigation: temperature, ensembling, and verbalized uncertainty

Goal. Compare post-hoc and prompting-based mitigations for overconfidence and quantify their accuracy/calibration trade-offs.

Approach. Implement three mitigations on a shared evaluation suite: (a) temperature scaling fit on a held-out set [2]; (b) deep-ensemble or sampling-based uncertainty, aggregating multiple stochastic decodes and using self-consistency as a confidence proxy [5, 7]; and (c) explicit prompting for calibrated verbalized confidence. Measure ECE, selective accuracy, and the cost in latency/compute for each.

Deliverables.

- A unified benchmark harness applying all three mitigations.
- Before/after reliability diagrams and a cost–benefit table.
- A recommendation note on which mitigation suits latency-constrained vs. accuracy-critical deployments.

Project 4 — Training dynamics of overconfidence

Goal. Trace how confidence miscalibration emerges during training or fine-tuning, connecting it to loss-surface and memorization phenomena.

Approach. Fine-tune a small model while checkpointing frequently. At each checkpoint, measure accuracy and ECE, and track whether the model becomes confident on examples it later gets wrong. Relate observations to the tendency of networks to fit noisy or memorized data with high confidence [8] and to over-sharpening under cross-entropy. If using preference fine-tuning, test whether reward optimization increases assertiveness independent of correctness [6].

Deliverables.

- Trajectory plots of accuracy and ECE across training steps.
- An analysis isolating when confidence decouples from correctness.
- A report linking the empirical dynamics to at least one theoretical account of overconfidence.

Submission notes

Each project must ship with runnable code, fixed random seeds, a data manifest, and a self-contained report. Grading weights reproducibility and the clarity of the causal story linking an *internal* mechanism to the observed symptom of confident errors, not benchmark leaderboard position.

References

[1] G. Alain and Y. Bengio. Understanding intermediate layers using linear classifier probes. *International Conference on Learning Representations (Workshop)*, 2017.

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[3] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023.

[4] S. Kadavath, T. Conerly, A. Askell, T. Henighan, D. Drain, E. Perez, N. Schiefer, Z. Hatfield-Dodds, N. DasSarma, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

[5] B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[6] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

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[8] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals. Understanding deep learning (still) requires rethinking generalization. *Communications of the ACM*, 64(3):107–115, 2021.